

Quiz 1 – Kinematics

Problem 1 (25 pts)

It is one-dimensional motion along x-axis. Figure on the right shows change in an acceleration with time. Find the mean velocity averaged over time interval $[2\text{ s}, 10\text{ s}]$, if the initial velocity equals to -8 m/s ?

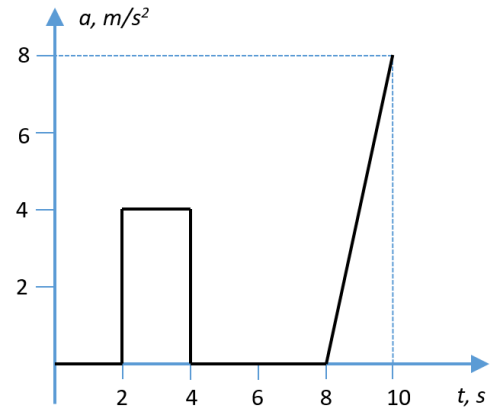


Figure for Problem 1

Problem 2 (25 pts)

The tram starts moving with the constant acceleration of 0.3 m/s^2 . Find the minimum passenger's velocity, when the passenger could wait for 8 s and then catch up the tram by running after.

Problem 3 (25 pts)

Two balls were thrown from the ground at angles of 30° and 45° to the horizon. The ranges R of the balls were the same. Find the ratio of their maximum heights during the flights. Neglect air resistance.

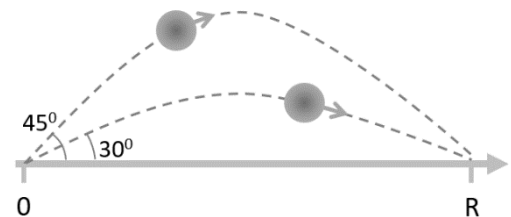


Figure for Problem 3

Problem 4 (25 pts)

A particle moves in uniform circular motion over a horizontal xy plane. At one instant, it moves through the point with coordinates $[-10\text{ m}, 10\text{ m}]$ with velocity of $(-20\hat{i} + 0\hat{j})\text{ m/s}$ and acceleration of $(0\hat{i} + 25\hat{j})\text{ m/s}^2$. Find the position of the particle at moment, when its velocity equals to $(20\hat{i} + 0\hat{j})\text{ m/s}$.